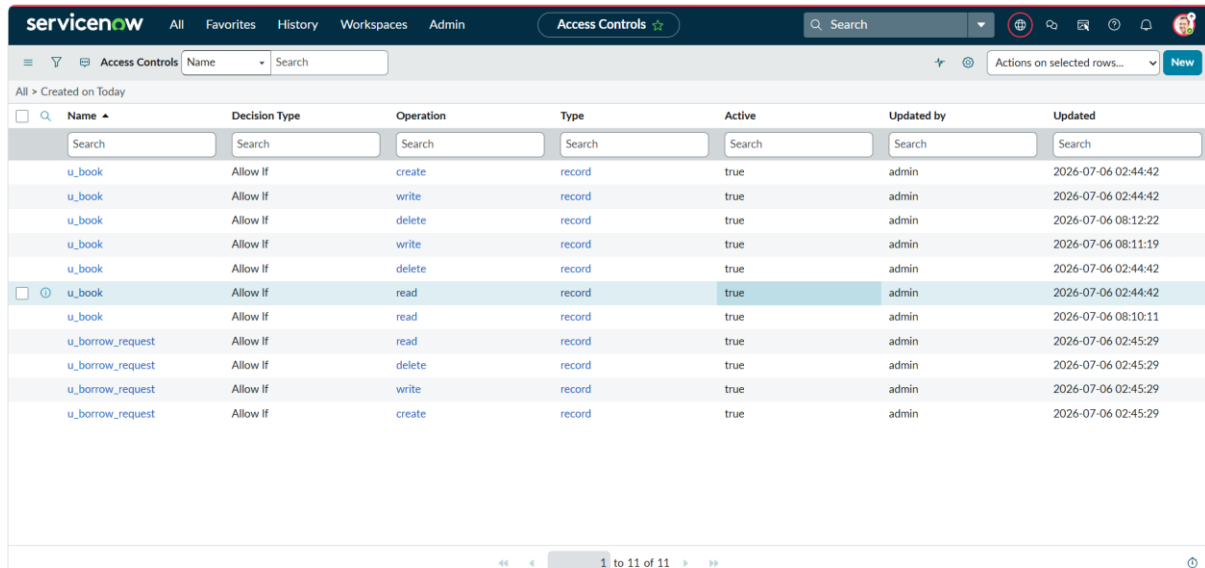


Data Migration, Testing & Security

1)Access Control for Book Table

The Access Control List (ACL) for the Book table is implemented to provide secure, role-based access to library records. Separate ACLs are created for the Read, Write, and Delete operations. The Read permission is granted to both Student and Librarian roles, allowing them to view available book details. The Write and Delete permissions are restricted to the Librarian role, ensuring that only authorized library staff can modify or remove book records. This security mechanism protects the integrity of book information and prevents unauthorized users from making changes to the library database.



The screenshot displays the ServiceNow 'Access Controls' page. The table lists ACLs for the 'u_book' table, showing various operations like create, write, delete, and read. The 'u_book' row is highlighted, showing 'Allow If' decision type, 'read' operation, 'record' type, and 'true' active status. The 'Updated by' column shows 'admin' and the 'Updated' column shows '2026-07-06 02:44:42'.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_book	Allow If	create	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	write	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	delete	record	true	admin	2026-07-06 08:12:22
u_book	Allow If	write	record	true	admin	2026-07-06 08:11:19
u_book	Allow If	delete	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	read	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	read	record	true	admin	2026-07-06 08:10:11
u_borrow_request	Allow If	read	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	write	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	create	record	true	admin	2026-07-06 02:45:29

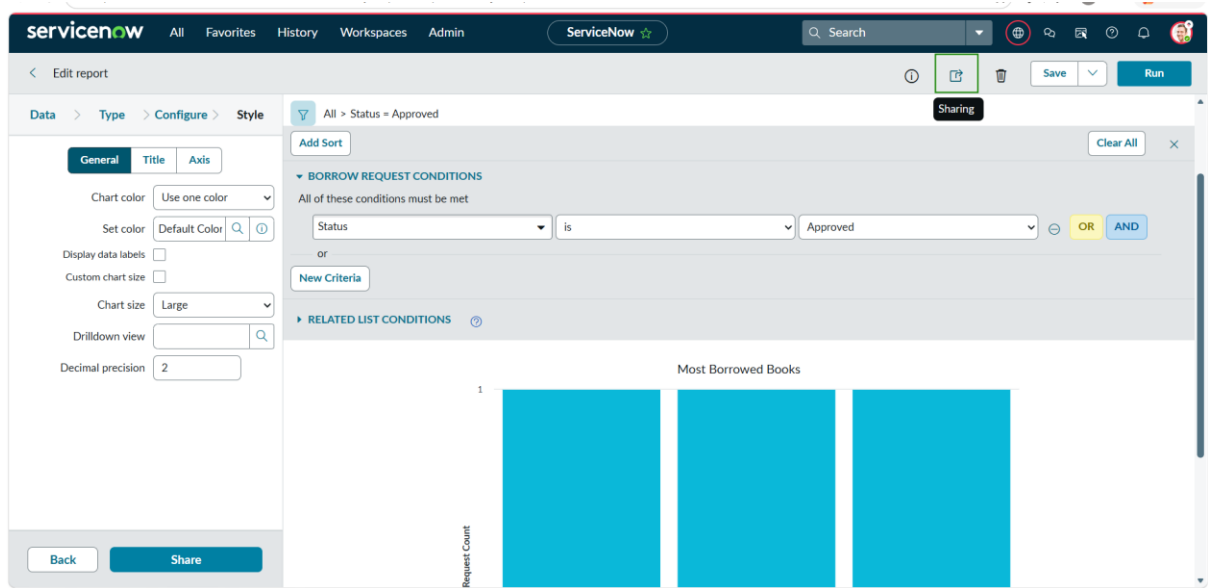
2)Access Control for Borrow Request Table

The Borrow Request table is secured using Access Control Lists (ACLs) to manage user permissions based on their roles. Students are provided with Create and Read access, enabling them to submit new borrow requests and view their request details. Librarians are granted Read, Write, and Delete permissions, allowing them to review, approve, reject, update, or remove borrow requests as required. This role-based access control ensures that students cannot modify approved requests while librarians maintain full control over the borrowing workflow, thereby improving application security and maintaining data consistency.

servicenow All Favorites Access Controls																																																																																																							
Access Controls Name Search Actions on selected rows... New																																																																																																							
All > Created on Today																																																																																																							
<table> <tr> <th></th><th>Name ▲</th><th>Decision Type</th><th>Operation</th><th>Type</th><th>Active</th><th>Updated by</th><th>Updated</th></tr> <tr> <td>☐</td><td>u_book</td><td>Allow If</td><td>delete</td><td>record</td><td>true</td><td>admin</td><td>02:44:42</td></tr> <tr> <td>☐</td><td>u_book</td><td>Allow If</td><td>read</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 02:44:42</td></tr> <tr> <td>☐</td><td>u_book</td><td>Allow If</td><td>read</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 08:10:11</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>write</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 08:47:10</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>read</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 02:45:29</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>delete</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 02:45:29</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>create</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 08:43:13</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>delete</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 08:54:09</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>read</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 08:44:39</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>write</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 02:45:29</td></tr> <tr> <td>☐</td><td>u_borrow_request</td><td>Allow If</td><td>create</td><td>record</td><td>true</td><td>admin</td><td>2026-07-06 02:45:29</td></tr> </table>									Name ▲	Decision Type	Operation	Type	Active	Updated by	Updated	☐	u_book	Allow If	delete	record	true	admin	02:44:42	☐	u_book	Allow If	read	record	true	admin	2026-07-06 02:44:42	☐	u_book	Allow If	read	record	true	admin	2026-07-06 08:10:11	☐	u_borrow_request	Allow If	write	record	true	admin	2026-07-06 08:47:10	☐	u_borrow_request	Allow If	read	record	true	admin	2026-07-06 02:45:29	☐	u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 02:45:29	☐	u_borrow_request	Allow If	create	record	true	admin	2026-07-06 08:43:13	☐	u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 08:54:09	☐	u_borrow_request	Allow If	read	record	true	admin	2026-07-06 08:44:39	☐	u_borrow_request	Allow If	write	record	true	admin	2026-07-06 02:45:29	☐	u_borrow_request	Allow If	create	record	true	admin	2026-07-06 02:45:29
	Name ▲	Decision Type	Operation	Type	Active	Updated by	Updated																																																																																																
☐	u_book	Allow If	delete	record	true	admin	02:44:42																																																																																																
☐	u_book	Allow If	read	record	true	admin	2026-07-06 02:44:42																																																																																																
☐	u_book	Allow If	read	record	true	admin	2026-07-06 08:10:11																																																																																																
☐	u_borrow_request	Allow If	write	record	true	admin	2026-07-06 08:47:10																																																																																																
☐	u_borrow_request	Allow If	read	record	true	admin	2026-07-06 02:45:29																																																																																																
☐	u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 02:45:29																																																																																																
☐	u_borrow_request	Allow If	create	record	true	admin	2026-07-06 08:43:13																																																																																																
☐	u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 08:54:09																																																																																																
☐	u_borrow_request	Allow If	read	record	true	admin	2026-07-06 08:44:39																																																																																																
☐	u_borrow_request	Allow If	write	record	true	admin	2026-07-06 02:45:29																																																																																																
☐	u_borrow_request	Allow If	create	record	true	admin	2026-07-06 02:45:29																																																																																																

3)Create Report

A report titled "Most Borrowed Books" is created using the Borrow Request table to analyze the borrowing activity within the library management system. The report uses a Bar Chart to display the number of approved borrow requests grouped by the Book field. A filter is applied to include only records with the Approved status so that only successfully issued books are counted. This report helps librarians identify the most frequently borrowed books, monitor library usage trends, and make informed decisions regarding book procurement and inventory management. The graphical representation provides an easy-to-understand overview of borrowing statistics and supports effective library administration.



Most Borrowed Books

Page 1

Report Title: Most Borrowed Books
Run Date and Time: 2026-07-06 09:33:28 Pacific Daylight Time
Run by: System Administrator
Table name: u_borrow_request
Query Condition: Status = Approved
Group by: Book

